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Connector with movable collar for selectively displaying indicia

Abstract

A female quick connector member for making fluid-tight connections includes a body having indicia and an axial opening for receiving and lockingly engaging a male quick connector member, a collar positioned over the body, a latch for holding the collar in a first position that conceals the indicia prior to coupling of the male and female connector members, a spring for urging the collar to a second position in which the indicia is fully exposed when the latch is released, a release member engageable by the male connector member and configured to release the latch when the male connector member is lockingly engaged with the female connector body.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) This disclosure generally relates to quick connector couplings providing fluid-tight connections in pipe and tubing fluid conduit systems, and more particularly to quick connector couplings that include means for displaying indicia around the connector in multiple places providing a visual indication for a fluid-tight connection that has been completed between quick connector members.

BACKGROUND OF THE DISCLOSURE

(2) There is a recognized need for providing connection verification indicating a fluid tight connection between a male quick connector member and a female quick connector member. Relatively simple mechanism for accomplishing this task have involved verification indicia that become viewable through a window or cutout in one of the connector members when a fluid tight connection has been achieved. A disadvantage with such verification mechanism is the need for proper alignment between the connector members to allow obstruction-free viewing of the indicia. Such disadvantage can be overcome by providing keyed or splined connector members that only allow aligned assembly. However, such alignment features make assembly more difficult.

(3) It is desirable to provide a verification mechanism that exposes indicia on the female quick connector member without regard to angular alignment of the connector member around an axis

coincident with the direction of insertion of the male connector member into the female connector member and exposing the indicia in multiple areas around/across the surface of the female connector.

SUMMARY OF THE DISCLOSURE

(4) The disclosed female quick connector member includes a body having indicia on an outer surface thereof and an axial opening for receiving and lockingly engaging a male quick connector member, a collar positioned over and axially movable along the body of the female quick connector member, a latch for holding the collar in a first position that conceals the indicia, a spring for urging the collar to a second position in which the indicia is fully exposed when the latch is released, and a release member engageable by a male connector member inserted into the axial opening and configured to release the latch when a male connector member is lockingly engaged with the female connector body, whereby the spring urges the collar to the second position fully exposing the indicia.

(5) In certain embodiments, the latch is a cantilevered section of the collar having a projecting detent that can engage a radial opening through the wall of the female connector body to hold the collar in the first position before insertion of the male connector member.

(6) In certain other embodiments, the female connector member may further comprise a retainer that resiliently deforms radially upon insertion of a male connector member and that holds the male connector member and female connector member together after complete insertion.

(7) In certain embodiments, the release member is a pin extending through a radial opening in the wall of the female connector body. The pin has a proximal end projecting into the axial opening of the female connector body and a distal end engageable with the latch, whereby complete insertion of the male connector member causes engagement with the pin and release of the latch to allow the spring to urge the collar into the second position in which the indicia are exposed.

(8) In certain embodiments, when the collar is in the second position to expose the indicia, the projecting detent of the cantilever section engages on top of the retainer enclosing it completely, thereby preventing the retainer from reopening, providing a secondary lock on the already connected male and female connector members.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a side view of a female quick connector member in accordance with this disclosure.

(2) FIG. **2** is a side view of the female quick connector member shown in FIG. **1** fully connected with a male quick connector member.

(3) FIG. **3** is a cross-sectional view of the male quick connector member partially inserted into the female quick connector member along a cutting plane including the axis.

(4) FIG. **4** is a cross-sectional view of the male quick connector member fully inserted into the female quick connector member along a cutting plane including the axis.

(5) FIG. **5** is a cross-sectional view of a coupled male-female assembly employing a plastic retainer along a cutting plane perpendicular to the axis.

(6) FIG. **6** is a cross-sectional view of a couple male-female assembly employing a metal retainer along a cutting plane perpendicular to the axis.

DETAILED DESCRIPTION

(7) Shown in FIG. **1** is a female quick connector member **10** having a body **12**. Indicia **14** is provided on an outer surface of a wall of body **12** in multiple areas. Indicia **14** is fully, or at least partially covered by a collar **16** (as shown in FIG. **1**) before connection with a male quick connector member **20** (FIG. **2**). As shown in FIG. **3**, female connector member **10** has an axial opening **22** for receiving an end **24** of male connector member **20**. Collar **16** can be fit over the

outer surface of body **12**.

(8) A first fluid conduit means such as a tube or pipe (not shown) is connected to the exposed end **26** of male connector member **20**, and a second fluid conduit means, which may be the same as or different from the first fluid conduit means, is connected to the end **28** of female connector member **10**. Typically, the conduit means are connected to the respective connector members before the connector members are joined to each other.

(9) A latch **30** is provided to hold collar **16** in a first position that conceals at least a portion of indicia **14** prior to joining of the male connector member **20** to the female connector member **10**. Latch **30** can be any device that locks collar **16** in the first position until male connector member **20** is fully inserted into axial opening **22**, and then, after full insertion, releases collar **16**, allowing it to be urged into a second position relative to body **12** by a spring **32**, whereby indicia **14** is fully exposed (as show in FIG. **2**).

(10) In the illustrated embodiments of FIGS. **1** through **4**, the latch **30** is a cantilevered section of collar **16** having a protruding catch or detent **34** that is resiliently biased into engagement with a radial opening extending through the wall of body **12**. A portion of the radial opening is occupied by detent **34** prior to connecting the female and male connectors together. The remaining portion of the radial opening is occupied by a release member or pin **36** that extends from detent **34** into opening **22**. During assembly of the quick connection, as male member **20** nears full insertion into axial opening **22**, an outer wall of member **20** contacts a proximal end of pin **36** opposite the distal end that is engaging detent **34**, and upon full insertion of member **20**, pin **36** and detent **34** are urged in a radially outward direction, unlocking collar **16** from body **12** and allowing spring **32** to force collar **16** to move from the first position (FIGS. **1** and **3**) to the second position (FIGS. **2** and **4**) in which indicia **14** are fully exposed.

(11) While collar **16** can be locked into a first position that partially exposes indicia **14**, it is desirable that a portion of indicia **14** is concealed in the first position to prevent reading of the indicia by a machine or human, and only upon full insertion of the member **20** and complete connection between members **10** and **20** is indicia **14** readable.

(12) Indicia **14** may include human readable characters such as letters, numbers, color, and/or symbols. Indicia **14** may include only machine-readable codes such as QR codes, bar codes or the like. Indicia **14** may include combinations of machine-readable codes, human readable characters, and color codes. Such indicia may be applied to body **12** as pre-printed stickers, or they can be laser etched, printed, painted, engraved, embossed, molded into, or otherwise applied. Multiple indicia can be circumferentially spaced apart along the outer surface of body **12** to allow machine readability from any angle or perspective.

(13) Spring **32** can be a coil spring, a helical spring, a leaf spring, a wave spring, or the like, made of a resiliently deformable metal (e.g., stainless steel) or other elastic material.

(14) A preferred embodiment includes two latches. However, generally any number of latches and associated release members may be used. Similarly, while a preferred embodiment includes four (4) uniformly angularly spaced apart springs, any number of springs may be used.

(15) A metal or plastic retainer **38** can be provided in a slot **39** of member **10** to help hold male connector member **20** and female connector member **10** together after complete insertion of the member **20** into member **10**. Retainer **38** is radially deformed outwardly by an outwardly facing ramped surface **40** of an enlarged (e.g., frustoconical) section (FIG. **3**) of male connector member **20** during insertion of member **20** into axial opening **22** (FIG. **4**). Upon full insertion of male connector member **20** into axial opening **22**, retainer **38** rebounds into a groove **42**, defined by the male connector member, helping to lock the members **10** and **20** together.

(16) As shown in FIG. **4**, latch **30** can be configured to cover and engage retainer **38** to provide a secondary lock for holding collar **16** in the second position to maintain exposure of indicia **14**.

(17) As shown in FIG. **5**, retainer **38** has an outwardly projecting yoke **44** to which pressure can be applied to release latches **30** to facilitate separation of members **10** and **20**.

(18) FIG. **6** shows an alternative metal (e.g., steel) retainer **38′** that has a yoke **46** that can be pulled radially outwardly to facilitate separation of members **10** and **20**.
(19) The above description is intended to be illustrative, not restrictive. The scope of the invention should be determined with reference to the appended claims along with the full scope of equivalents. It is anticipated and intended that future developments will occur in the art, and that the disclosed devices, kits and methods will be incorporated into such future embodiments. Thus, the invention is capable of modification and variation and is limited only by the following claims.

Claims

1. A female quick connector member, comprising: a female connector body having an outer wall, the female connector body having an axial opening adapted to receive and lockingly engage a male connector member; indicia on the outer wall of the female connector body; a collar positioned over the outer surface of the female connector body for axial movement along the outer wall; a latch for holding the collar in a first position that at least partially conceals the indicia; a spring for urging the collar toward a second position exposing the indicia when the latch is released; and a release member extending from an interior wall of the axial opening, the release member engageable by a male connector member inserted into the axial opening, the release member configured to release the latch when the male connector member is lockingly engaged with the female connector member, whereby the spring urges the collar to the second position exposing the indicia; a radially deformable retainer located in a slot defined in the female connector body that is configured to deflect radially outwardly during insertion of the male connector member and rebound into a groove on the male connector member to lock the male and female connector members together; and wherein the latch is configured to cover and engage the retainer upon complete insertion of the male connector member into the female connector member to provide a secondary lock.
2. The connector member of claim 1, wherein the latch is a cantilevered section of the collar having a catch that projects into a radial opening through the wall of the female connector body before insertion of the male connector member.
3. The connector member of claim 1, wherein the spring is a coil spring, a leaf spring or a wave spring.
4. The connector member of claim 1, further comprising a retainer that deforms radially upon insertion of the male connector member and that holds the male connector member and female connector member together after complete insertion into the female connector member.
5. The connector member of claim 4, wherein the latch is a cantilevered section of the collar having a catch that projects into a radial opening through the wall of the female connector body before insertion of the male connector member.
6. The connector member of claim 1, wherein the release member is a pin extending through a radial opening in the wall of the female connector body, the release member having a proximal end projecting inwardly into the opening of the female connector body, the proximal end of the release member engageable with the male connector member upon complete insertion of the male connector member into the female connector, the release member having a distal end opposite the proximal end, the distal end engageable with the latch, whereby complete insertion of the male connector member into the female connector member causes release of the latch, and urging of the collar by the spring into the second position to expose the indicia.
7. The connector member of claim 1, wherein the female connector body has multiple indicia on the outer surface of the wall.
8. The connector member of claim 1, wherein the connector member has a plurality of latches for holding the collar in the first position.
9. The connector member of claim 1, wherein the connector member has a plurality of springs urging the collar toward the second position.

10. The connector member of claim 1, wherein the indicia includes a machine-readable code.
11. The connector member of claim 10, wherein the machine-readable code is a QR code or a bar code.
12. The connector member of claim 1, wherein the indicia includes words, numbers and/or symbols that can be read by a human.